

SPINEL User Documentation

This software is written entirely in Python and has only been tested to run on macOS.

Installation

1. Ensure you have a working Python distribution installed on your machine.
2. Navigate to the following link and download a copy of the entire repository to a suitable location on your machine (a .zip file can be downloaded using the dropdown arrow next to the green “Code” button):

https://github.com/jdmchardy/SPINEL_XRD

3. Open a terminal window and run (if you have not installed poetry previously on your machine):

```
pip install poetry (or on windows: py -m pip install poetry)
```

4. Verify the installation with:

```
poetry --version
```

5. In the terminal, navigate to the root folder where you saved a copy of the SPINEL Github repository on your machine.

6. Confirm you are in the right directory by running ls (dir on windows). You should see the following contents:

```
logo.png
pyproject.toml
requirements.txt
poetry.lock
README.md
spinel.py
SPINEL_user_documentation.pdf
example_files/
```

7. Install the necessary dependencies:

```
poetry install
```

8. Start up the program using:

```
poetry run streamlit run spinel.py
```

Overview

SPINEL computes the deviatoric strain normal to a set of diffraction planes resulting from a macroscopic uniaxial stress field for cubic, hexagonal and tetragonal crystal systems.

Details of the theoretical models can be found in McHardy et al. <https://doi.org/10.1063/5.0328151> or Uchida et al. [<https://doi.org/10.1063/1.362920>] and Funamori et al. [<https://doi.org/10.1063/1.365792>].

Startup

1. Successful startup will launch the following window in your browser



Upload Files ↔

Elastic and hkl csv



Drag and drop file here

Limit 200MB per file • CSV

Browse files

2. Upload an example .csv file from the example_files folder or one you have modified for your simulation. The meta data will be imported and displayed on screen.

The imported values can be adjusted and the hkl reflections included can be toggled on and off.

Reflections/Intensities	Material	Elastic	Stress	Computation	Preferred Orientation				
☒ hkl = (1, 1, 0)	104.00 - +	Symmetry: cubic	Lattice a (Å) 3.1490 - +	alpha (deg) 90.000 - +	c11 441.00 - +	σ_{11} -1.000 - +	σ_{12} 0.000 - +	☒ Include broadening	☒ Preferred Orientation
☒ hkl = (2, 0, 0)	38.00 - +	Wavelength (Å) 0.4832 - +	Lattice b (Å) 3.1490 - +	beta (deg) 90.000 - +	c12 172.00 - +	σ_{22} -1.000 - +	σ_{13} 0.000 - +	ψ steps (output/plot) 60 - +	PO Model: March-Dollas
☒ hkl = (2, 1, 1)	42.00 - +	Chi angle (deg) 0.000 - +	Lattice c (Å) 3.1490 - +	gamma (deg) 90.000 - +	c44 122.00 - +	σ_{33} 2.000 - +	σ_{23} 0.000 - +	ϕ steps (integration) 18 - +	POD hkl 110
☒ hkl = (3, 1, 0)	18.00 - +							α steps (integration) 18 - +	Baseline (between 0 and 1) 0.00 - +
☒ hkl = (2, 2, 2)	5.00 - +							Total points: 19,440 (60 ψ × 18 ϕ × 18 α)	R 0.200 - +
								Gaussian FWHM 0.100 - +	tau (deg) 0.0 - +
									omega (deg) 0.0 - +
									weight 1.0 - +

Simulation Options

1. Current permissible crystal symmetry options are:
 - cubic (space group #195-230),
 - hexagonal (space group #195-230),
 - tetragonal_A (space group #89-142),
 - tetragonal_B (space group #75-88)
 - orthorhombic (space group #16-74)
 - trigonal_A (space group #149-167)
2. chi angle (deg): the angle between compression axis and x-ray axis (0° = axial, 90° = radial)
3. elastic constants and stress component units - (GPa)
4. Diagonal (ii) and off-diagonal (ij) stress components.
 - a. The uniaxial stress component ($t = \sigma_{33} - \sigma_{11}$) is displayed below the stress parameters

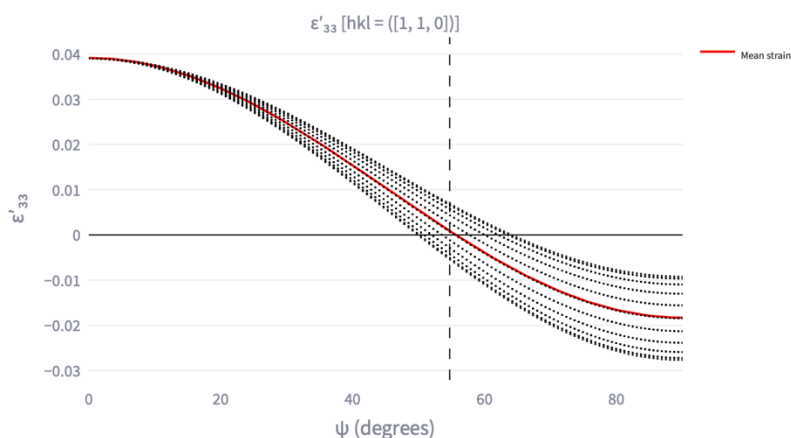
- Sampling steps for ψ , ϕ and α at which to evaluate the stress at (ψ ranges, between 0 to 90°, ϕ from 0 to 360° and α from -180 to 180°)
- Gaussian FWHM: the width of the peak shape contribution in the 1d integrated pattern added for each contributing crystallite orientation in 2th degrees.
- Include broadening: toggles the effects of broadening on the XRD and cake plots. With broadening off, all peaks have equal FWHM but are displaced from the ideal lattice sites based on the magnitude of t . The peak positions are computed from the mean strain across contributing orientations. In axial geometry ($\chi = 0$), broadening off yields results equivalent to the Singh formalism.
- Preferred orientation: toggles the scaling of contributions but the sample texture. Currently users can implement a March-Dollase fibre texture model. The hkl input under this tab set the diffraction normal reference vector. R-parameter controls the degree of texture. R=1 is an isotropic distribution. The angles τ and ω define the preferred direction relative to the principle stress axis. τ is the inclination and ω the rotation about the stress direction. Setting both values to zero aligns the stress and the preferred orientation axis.

Output

Once the simulations parameters are set, several calculations are possible:

1. ϵ - ψ Curves

Generates a series of plots (one per hkl reflection) showing the deviatoric strain distribution over all crystal orientations. The point density for these simulations is controlled by the sampling steps for ψ , ϕ and α .



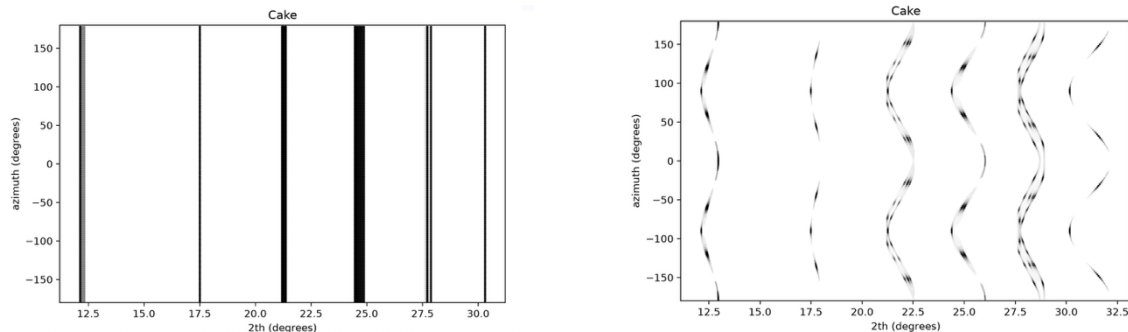
All calculated values are downloadable beneath the plots as .xlsx, .ods, or .csv files.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
1	hkl	h	k	l	strain_33	psi (degrees)	phi (degrees)	chi (degrees)	delta (degrees)	alpha (degrees)	d strain	2th	intensity	PO_intensity	Mean strain @ psi	Mean two_th @ psi	Mean l @ psi	Mean strain @ delta	Mean two_th @ delta	Mean l @ delta
2	110	1	1	0	0.038571734	173.77	0	90	-180	-180	2.14079	12.960	104	3.694	0.038	12.958	384.225	0.038	12.958	384.225
3	110	1	1	0	0.038570083	173.77	5	90	-180	-180	2.14080	12.960	104	3.694	0.038	12.958	384.225	0.038	12.958	384.225
4	110	1	1	0	0.03856518	173.77	10	90	-180	-180	2.14081	12.960	104	3.694	0.038	12.958	384.225	0.038	12.958	384.225
5	110	1	1	0	0.038557174	173.77	15	90	-180	-180	2.14082	12.960	104	3.694	0.038	12.958	384.225	0.038	12.958	384.225
6	110	1	1	0	0.038546307	173.77	20	90	-180	-180	2.14085	12.960	104	3.694	0.038	12.958	384.225	0.038	12.958	384.225
7	110	1	1	0	0.038532912	173.77	25	90	-180	-180	2.14088	12.959	104	3.694	0.038	12.958	384.225	0.038	12.958	384.225
8	110	1	1	0	0.038517393	173.77	30	90	-180	-180	2.14091	12.959	104	3.694	0.038	12.958	384.225	0.038	12.958	384.225
9	110	1	1	0	0.038500223	173.77	35	90	-180	-180	2.14095	12.959	104	3.694	0.038	12.958	384.225	0.038	12.958	384.225
10	110	1	1	0	0.038481924	173.77	40	90	-180	-180	2.14099	12.959	104	3.694	0.038	12.958	384.225	0.038	12.958	384.225
11	110	1	1	0	0.038463052	173.77	45	90	-180	-180	2.14103	12.958	104	3.694	0.038	12.958	384.225	0.038	12.958	384.225

Results are separated on different tabs by hkl and one row is given per (ϕ , ψ) pairing. Mean strain and mean 2th are averaged per each unique ψ -value or for each unique δ -value thus providing an average over the “Funamori” sampling or the average at each azimuth.

2. Cake plots

Iterates over azimuth (δ) and phi to evaluate only the (φ, ψ) pairings contributing to the x-ray diffraction (XRD) pattern. For each hkl, ψ is evaluated using the list of azimuth values, chi input, and theta (Bragg angle for each hkl). A 'cake' plot of azimuth vs. 2th is generated as well as individual strain vs 2th curves for each reflection. Again, calculation results may be downloaded as .xlsx, .ods or .csv files.

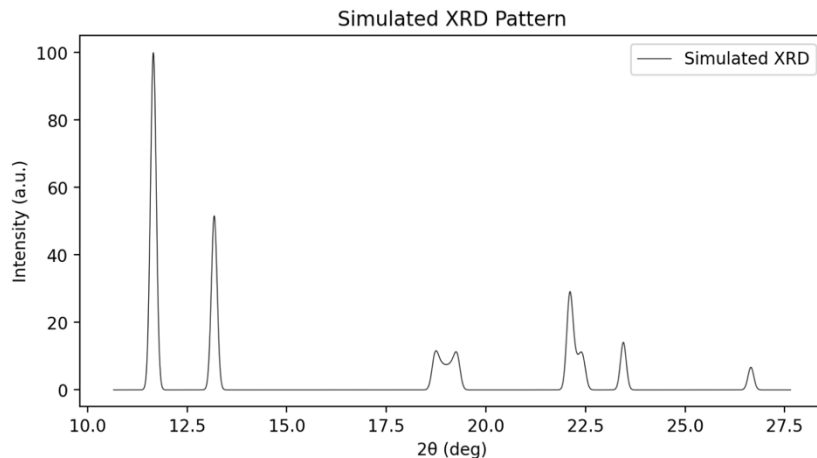


Generate 1D-XRD

Contributions from all incorporated orientations are combined to simulate an XRD pattern. The pattern is constructed by convoluting a Gaussian kernel with a histogram of weighted contributions in 2th. The contributions are weighted by the specified peak intensity and preferred orientation if toggled on. Simulated patterns can be downloaded as .xy files of 2th coordinate vs intensity.

Generate XRD patterns

Generate 1D-XRD



 Download simulated xy data (.xy)